Running your paper portfolio

The account marked to market every night, the risk report, two new refusals — and why a drawdown is a measurement, not a fault.

We grade decisions, not outcomes.

Education, not advice.

0:00 — HARVEST

The Monday routine, once

Round the room

What did the rule say for your instrument on Monday? What did you write? Did you ticket? Count: long and holding · long and ticketed · flat and waiting · warming.

Two questions

“Whose account is below where it started?” — most hands. “Whose is below its own highest value?” — every hand that holds anything.

Tonight

What the page now shows you every night, why the second question is the normal state of a running account, and what the rules do about it.

A rule you can hold for three weeks is a hobby; one you can hold through a 20% drawdown without touching it is a system.

What the page now shows

Figure	How	The test account
Account value	cash + Σ shares × the latest close the toolkit holds	`$3,893.02 + 18 × $331.90 = $9,867.22`
Open P&L	(latest close − entry) × shares	`(331.90 − 338.32) × 18 = −$115.56`
Realised P&L	closed positions, net of both tickets' costs	`$0.00 (after the 5 Nov close: +$166.47)`
High-water mark	the highest value the nightly report has seen; starts at the enrolled capital; only rises	`$10,000.00`
Drawdown	(high − value) ÷ high	`(10,000 − 9,867.22) ÷ 10,000 = 1.3%`
Risk used	Σ (position ÷ account) × instrument risk — week 8's arithmetic, backwards	`60.55% × 19.04% = 11.5% of the 12% budget`
Nearest stop	(latest close − stop) ÷ latest close	`(331.90 − 306.11) ÷ 331.90 = 7.8%`

Every figure is a line a student can check on a calculator from the page's own numbers. The report says the same thing in sentences — statements, never instructions.

THE REPORT

Four sentences, every night

On the 2026-11-03 close the account is worth $9,867.22: $3,893.02 cash and $5,974.20 in 1 open position — −1.3% since enrolment ($0.00 realised, −$115.56 open).

The high-water mark is $10,000.00; the account is 1.3% below it. At a 12% target a one-in-twenty year is about −20%; a run of years typically sees a drawdown of 20–30% from a high, one run in twenty 40% or more, and deeper over a decade: this is what the target was chosen to survive. Nothing in the rules changes because of it.

The open positions carry 11.5% a year of the 12% budget (AAPL 11.5%); the account runs 1 instrument, each sized at 12%.

The nearest stop is AAPL, 7.8% below the latest close.

Statements

Worth, cash, below the high, risk used, the nearest stop. No “should”, no “consider”. The decision is yours and goes in the journal.

Flags, not alarms

drawdown ≥ 10% · over the 12% budget · stop hit · more positions than instruments · no close stored — the same words on the page, in the lesson and here; each a sentence the student can verify, kept in the nightly row.

The habit

Monday: read the four sentences, quote one number in the journal. That is the whole of the risk layer from the student's side.

0:25 — DRAWDOWNS ARE NORMAL

From the one number the course chose

A 12% year is ordinary

The target is the account's volatility: it moves about 12% a year either way. Before any expected gain, half of all years end below where they started — and the rule's expected gain is small and slow (week 7: about a quarter of a unit of return per unit of risk).

One in twenty ≈ −20%

1.65 standard deviations of a 12% move is 19.8%. The account enrolled at $10,000 should expect, once in a career of such years, to see $8,000 — with the rules followed perfectly.

From a peak: deeper

Peaks and troughs do not fall on the calendar. A run of years at a 12% target typically sees a drawdown of 20–30% from a high; one run in twenty 40% or more; a decade deeper still. A trend rule's are longer than most — it earns in a few long runs and loses a little most of the rest of the time.

The size was set so that the worst of these are survivable. That is what a risk target is for — not to prevent the drawdown, but to make it one the account lives through.

Education, not advice.

The course's figures: a random walk at 12% a year, simulated (no fat tails) — a floor, not a ceiling. Carver's own table: the risk-target chapter (tutor, OI-40)

WHAT THE RULES DO ABOUT IT

Nothing — and what you do: write

The size was set for it

Week 8's 12% is the dial that makes a 20–30% drawdown survivable. Changing the size in the drawdown is changing the dial on the worst night to turn it.

The stop closes a position

Not the account. A close at or below the stop is a decision already made, one position at a time, with a ticket and a journal line. The account's drawdown has no closing rule.

You write

“8% below the high; the rule is still long; the stop is 6% away; I am doing nothing.” The most valuable sentence in the course. Week 7's flawed brain wants to act; the arithmetic is the restraint.

“Why not sell when it's down 10% and buy back lower?” — Because that is a second rule, and you would have to show it works. Write it as a rule, for everyone, next week; we will see what it would have done. The 10% note on the page is a statement, not a threshold.

Three changes, one sentence each

Size on the current value

The account that fell to $9,867.22 sizes its next position on $9,867.22; one that rose sizes on more. The 12% is a target for the account as it is. A week-8 position can be a little over or under tonight — the page says so; the rules don't trim it, the stop does its work.

Instruments, N — concentration

1 unless the tutor sets more (up to 4). Each position is sized at 12% ÷ N: 6% each for two, 4% for three, 3% for four — the account as a whole still carries about 12%. A ticket that would open position N + 1 is refused: close one first, or ask for more instruments.

The budget

Before an opening ticket: tonight's risk of the open positions + the new one's ≤ 12% (+1 point for rounding). The sizing rule keeps this true by construction; it bites when a share got wilder since entry, or N was lowered with positions open. It never trims what you hold.

Carver's full system raises the total for diversification between instruments (the instrument diversification multiplier). The course sets it to 1: two US large caps that fall together on the same bad day are not the diversification the multiplier assumes.

src/lib/tickets.ts — decision 20: current value, N instruments at 12% ÷ N, the budget; IDM = 1. /size takes N too, so the calculator and the ticket agree

THE EXIT CHECK

What the new refusals look like

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Concentration: your account runs 1 instrument and 1 position is open (AAPL). Close one first, or ask the tutor
to set more instruments — each position is then sized at 12% ÷ N, so adding a name halves, or cuts by a third,
the others' size next time they are opened.

Oversized. At 19.04% instrument risk and the 6% target (12% ÷ 2 instruments), an account worth $10,000.00
supports $3,151.26 of AAPL — 9 shares at $331.90. You asked for 18 ($5,974.20, 60% of the account, carrying
11.4% a year). The rule's size is the most you can hold; fewer is allowed.

Over the risk budget: the open positions carry 11.5% a year tonight and 29 shares of KO would add 5.8% — 17.4%
against the 12% target (tolerance 1 point). Fewer shares, or wait for a stop to do its work.   (the week-8
position at the full 12%, N raised to 2 — seven KO shares are accepted)
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Every number checkable

Account value, target, N, shares, what you asked, what it would carry. 10,000 × 6 ÷ 19.04 = 3,151.26; 18 × 331.90 ÷ 10,000 × 19.04 = 11.4. The refusal quotes the arithmetic.

Nothing is written

A refused ticket is answered, not stored. Closing tickets are never gated by N, the budget or the tier: a stop is a stop.

Still not advice

None of the three recommends anything. They say what the rule would hold tonight, and refuse what it would not.

The test account on the 3 Nov 2026 close (Apple $331.90, 19.04%; Coca-Cola $94.08, 21.11%) — pinned in src/lib/tickets.test.ts and sizing.test.ts

0:55 — SMALL CAPS, NOW

128 closes, and a Q

128 closes first

A small-cap instrument can be ticketed once its rule has 128 closes — early December for the first visited. Until then the signals page shows “N of 128 closes” and the ticket says so.

Q — quarantined

A one-day move of more than 40% in the adjusted close with no splits row within four days: a Q on the signals page with the reason, and no opening ticket until the tutor adds the row (a split, or factor 1 with a note when it was news).

Why

An unexplained halving looks to the rule like a crash and to the volatility estimator like a 50% day: a position sized on that number is a quarter of the right size, and a long opened after it is opened on a false signal. The rule only works on prices that mean the same thing all the way along.

Closing tickets are never gated. Mueller Industries' split is the reason MLI is not quarantined: the table explains the jump.

1:05 — LIVE

Your page, tonight

`[ course URL ]/tickets/<your code>`

`[ course URL ]/api/risk/<your code>`

1 · Read the four sentences of your report. Say them back to your pair in your own words.

2 · Check one figure by hand from the page's own numbers: account value, drawdown or risk used. If you disagree with the page, that is a row to find — tell the tutor.

3 · Write the Monday line now, quoting one number (“2.3% below the high”). If the rule is long and you hold nothing: an opening ticket, units blank. If the stop was hit: the close. Otherwise: nothing, in writing.

4 · One refusal on purpose: an opening ticket in a second name. Read the concentration sentence.

Monday: five minutes, with the report

Monday evening, in this order

/signals for your instrument → your page: the stop, the stamp, the report → the journal → (maybe) a ticket. The nightly rows at the bottom of the page are your account's history; a flag there is a statement, not an instruction.

1 · The routine, with a number

The Monday line quoting one figure from the report. 2 · Check the report by hand: value, drawdown, risk used for your own page; bring the arithmetic.

3 · Read · 4 · Bring a question

The risk-target chapter again, for drawdowns; Systematic Trading ch. 9 for the method, ch. 11 for the multiplier the course set to 1. And one question you would want to ask a tutor bot about the course, in writing.

Close with the sentence: we grade decisions, not outcomes.

GO DEEPER — FREE, FOREVER, OPTIONAL

Where the numbers come from

The report is one file; Carver's reasoning is two books; the arithmetic is on every page:

BOOK 2019 Carver, Leveraged Trading — the Starter System: the risk target and the account's year; running more than one instrument

BOOK 2015 Carver, Systematic Trading ch. 9 (volatility targeting: what a target implies for drawdowns) · ch. 11 (instrument weights and the diversification multiplier the course sets to 1)

CODE src/lib/risk.ts · src/lib/tickets.ts · src/jobs/risk.ts — the mark-to-market, the report's sentences, the pre-trade checks, the nightly pass; src/lib/risk.test.ts, tickets.test.ts and sizing.test.ts pin every figure on these slides

PAGE /tickets/<code> · /api/risk/<code> · /signals — your account, its report, and the Q that keeps a false signal out of it